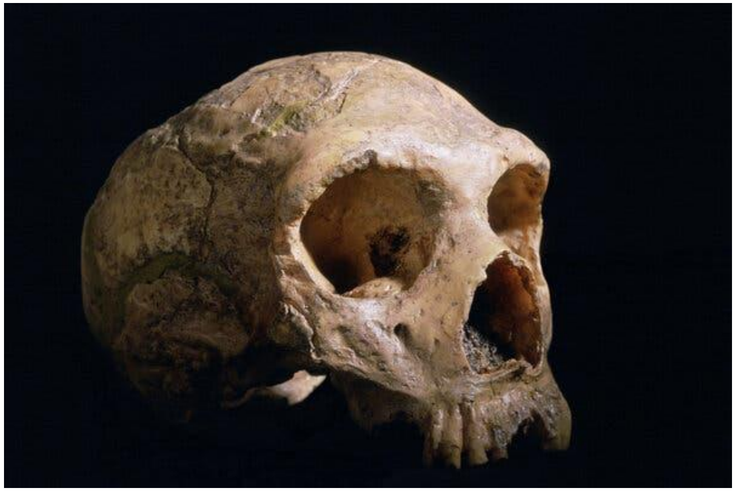
In the news…

['Happy evolutionary accident': How crayfish living underground may have gotten their bright colors]

A new study suggests the trait may have happened randomly and not as a survival tactic In a study published in July 2024, in the *Proceedings of the Royal Society B*, an ecologist at West Liberty University, and colleagues propose that the colorful hues of crayfish evolved not for a specific purpose—such as to attract mates or warn predators—but instead purely by chance.

[What Makes Your Brain Different From a Neanderthal's?]



Paleoanthropologists have found that the brains of our ancestors increased in size starting about two million years ago.

Scientists have discovered a mutation that increases the production of brain cells and seems to have set our ancestors apart from other hominins.

The Dynamic Genome

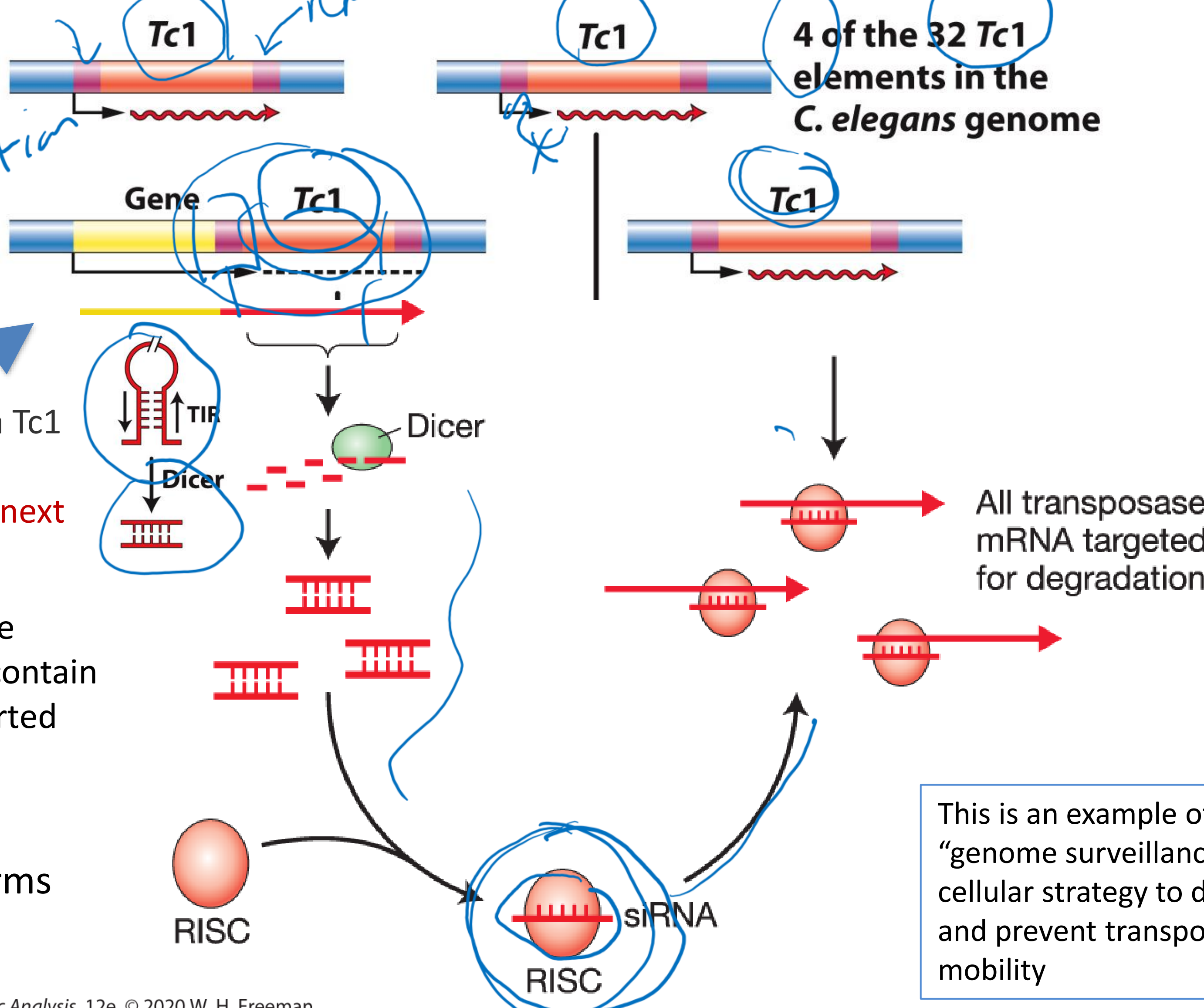
- Discovery of transposable elements through study of "unstable" alleles
- Impact of transposable elements on the genome
- Genomic surveillance mechanisms that limit transposon mobility

A single Tc1 can repress all Tc1 mobility in a cell

transposase gene

repeats

read-through transcription



“read-though” transcription from a Tc1 transposon that happened to insert next to a cellular gene.

Both ends of *Tc* 1 are transcribed. These contain 54-bp terminal inverted repeats (TIRS)…

… so the *Tc*1 RNA spontaneously forms dsRNA

This is an example of “genome surveillance” – a cellular strategy to detect and prevent transposon mobility

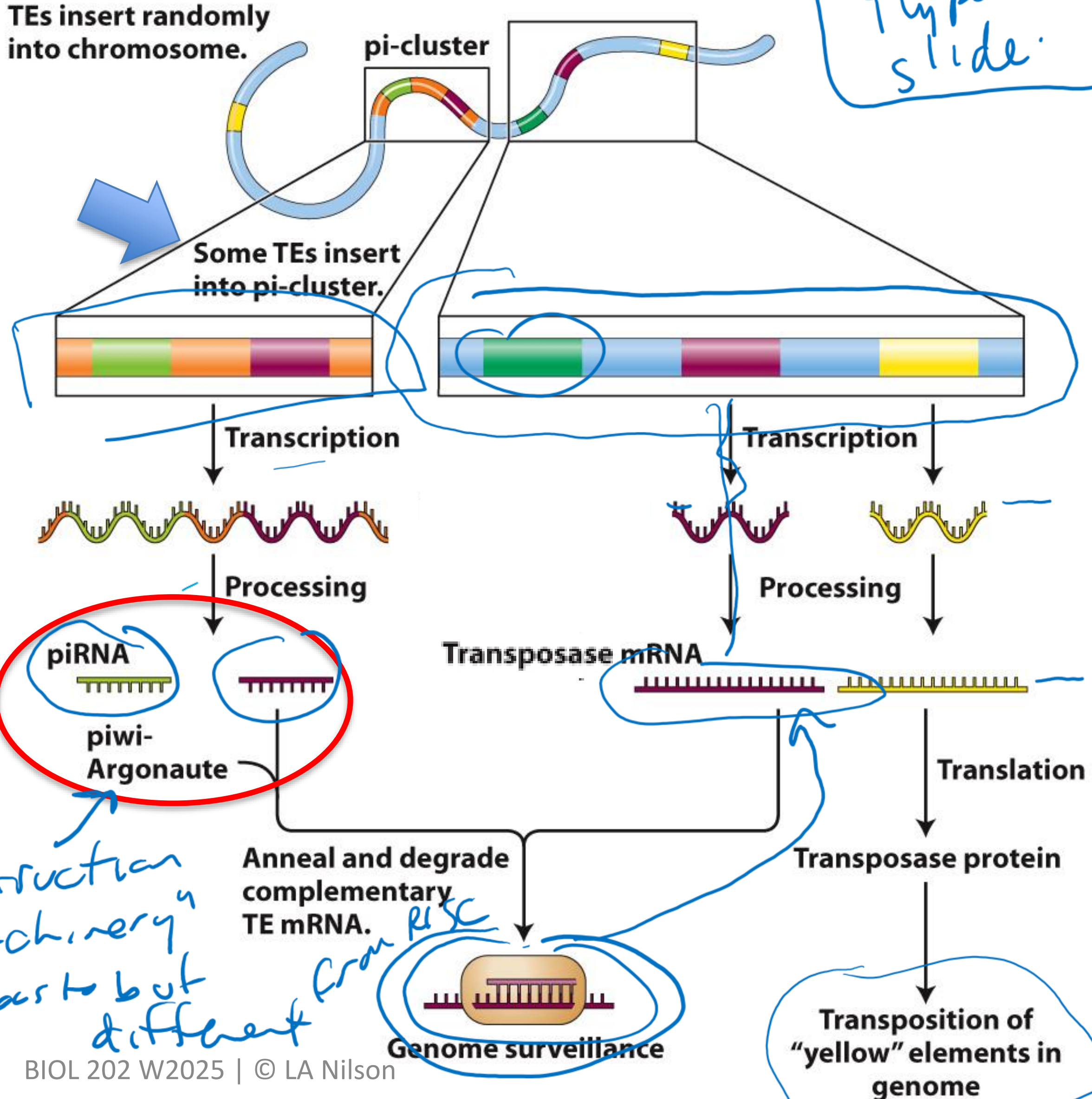
piRNAs repress transposon mobility in the germ lines of Drosophila and other animals

small RNAs

check Type 1 slide.

Like siRNAs, piRNAs are short single-stranded RNAs (26–30 nt in mammals) that interact with a protein complex that destroys complementary target mRNAs.

Unlike siRNAs, piRNAs do not start out as double-stranded RNA, but instead as parts of long mRNAs from a specific genome region called a pi-cluster.



"destruction machinery"
analogous to but different from RISC

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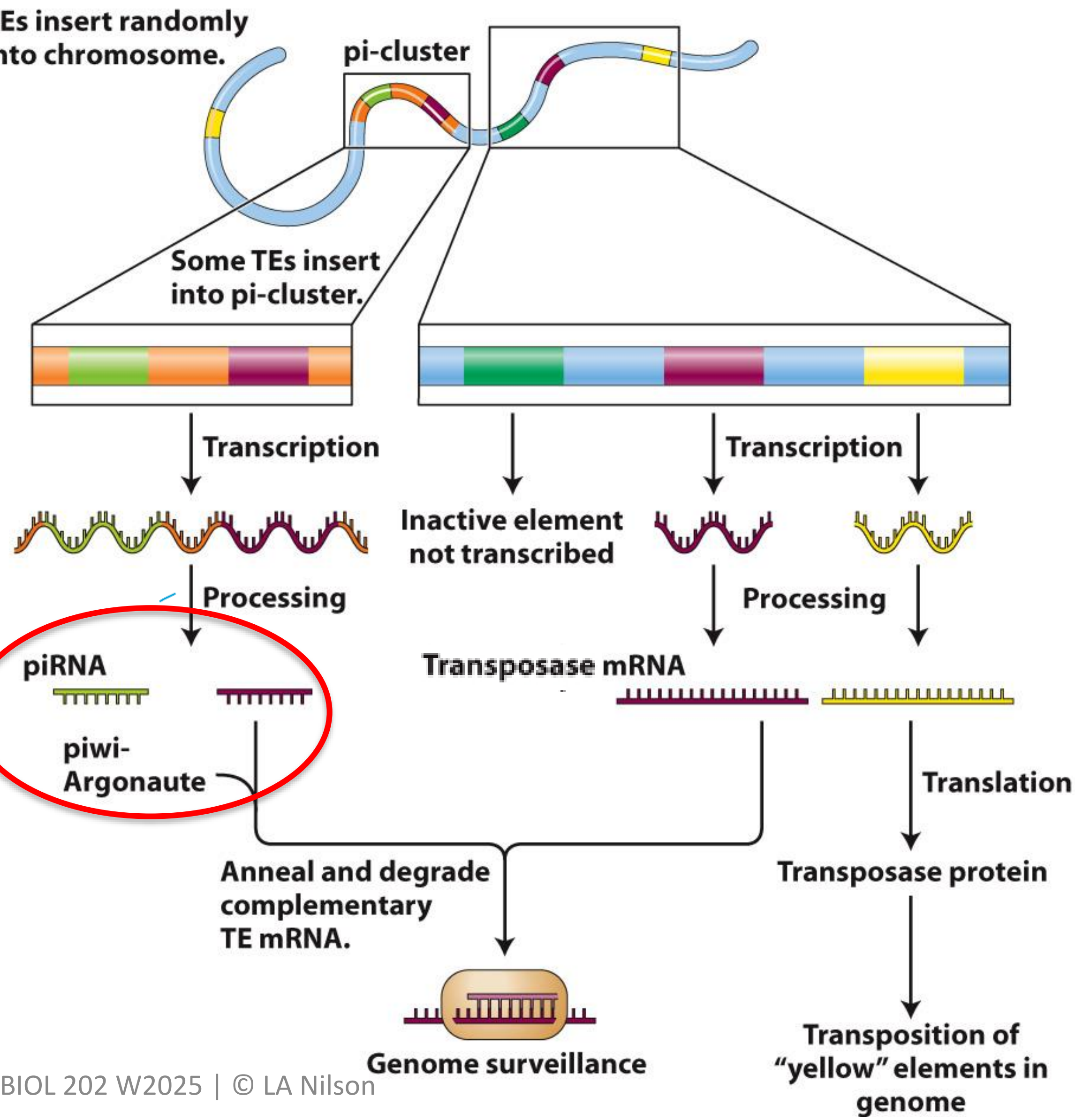
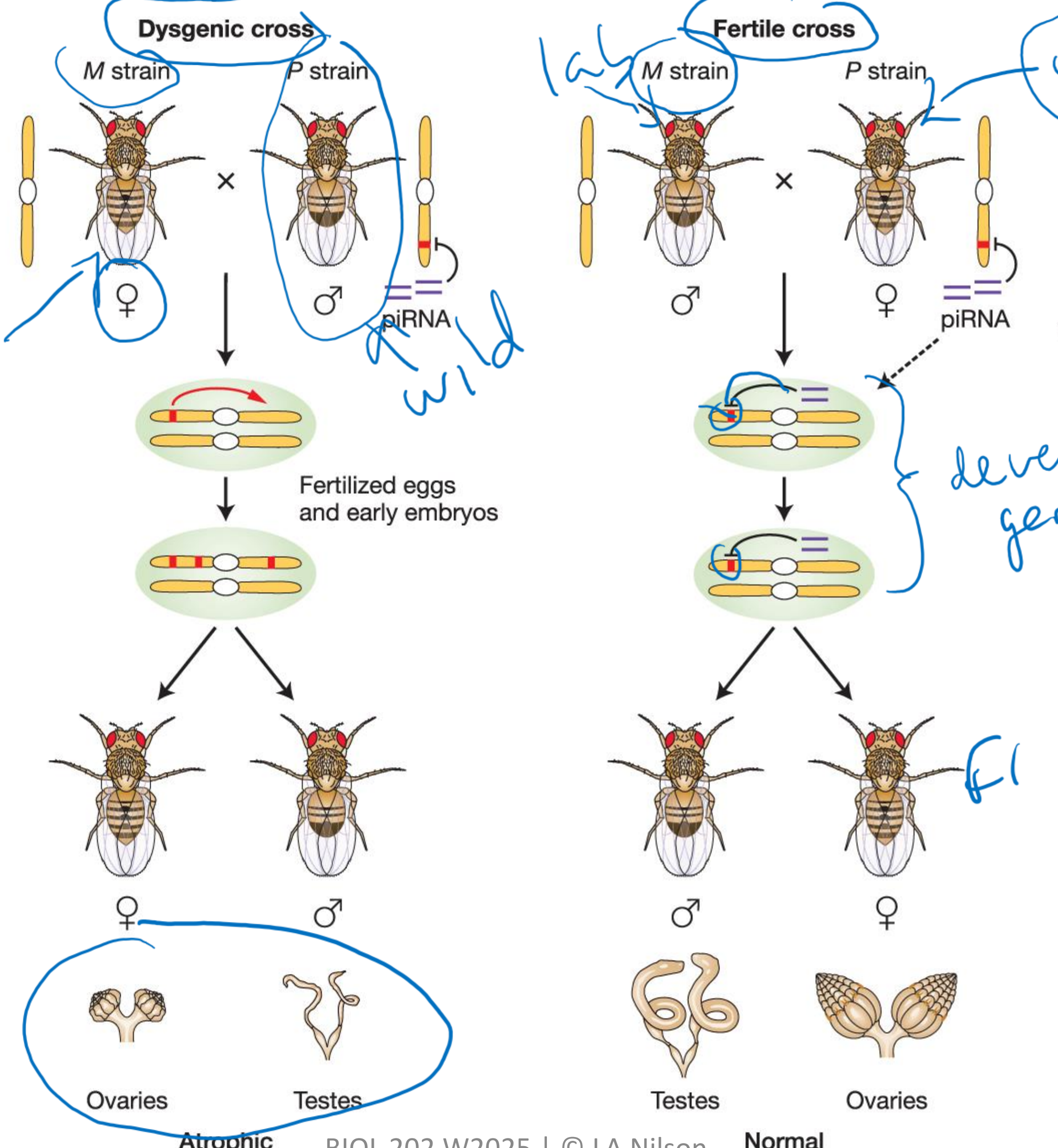


“Hybrid dysgenesis” explained:
piRNAs are present in P strain but not M strain

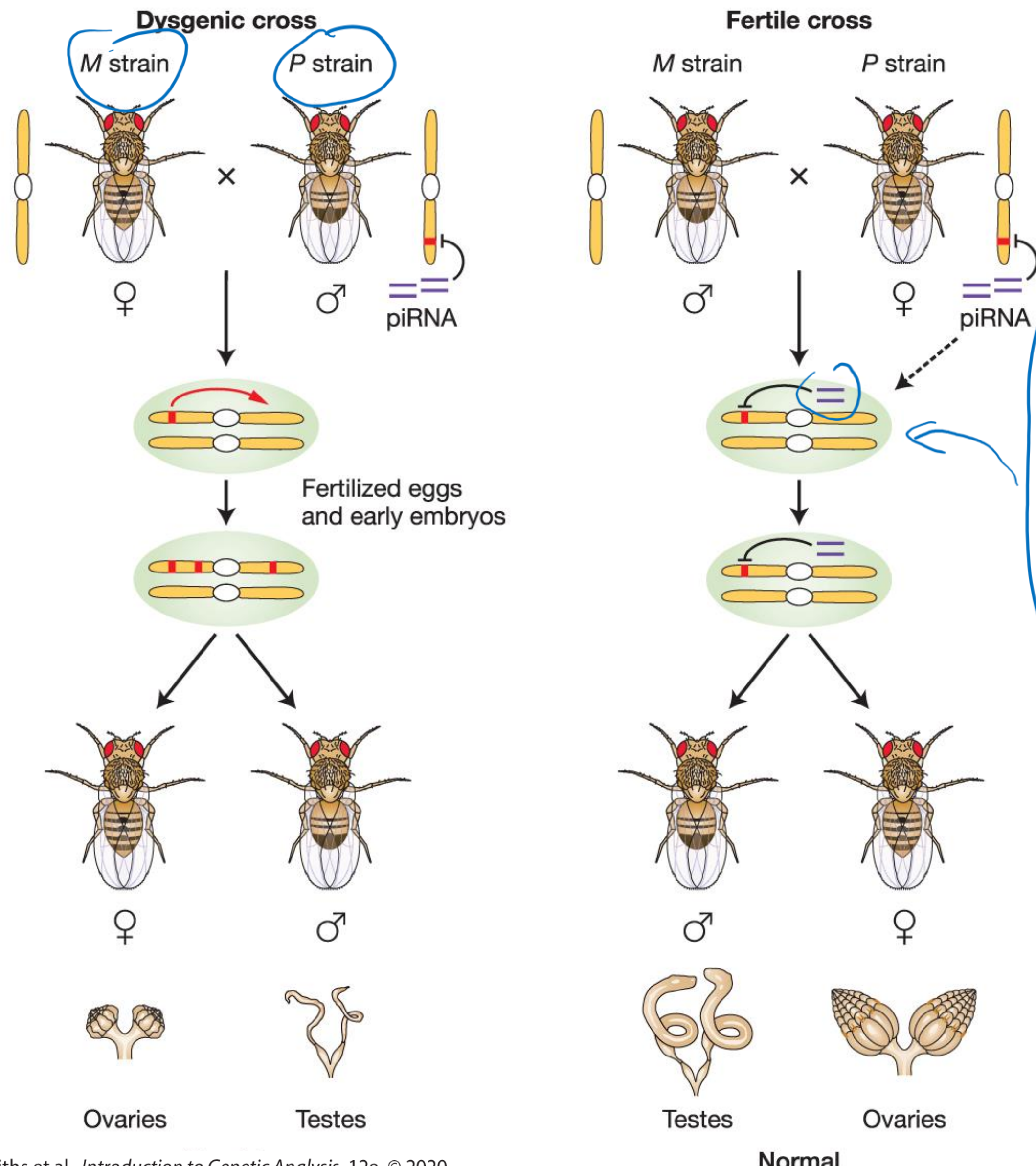
lab

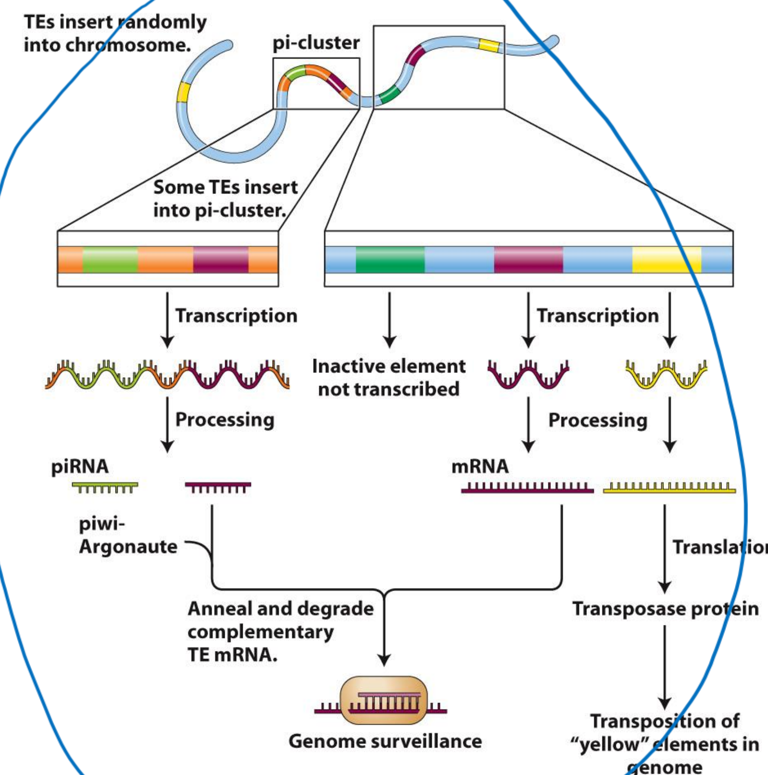


Lab strain female (M strain) has no P elements in her pi cluster and therefore no piRNAs in her eggs. P elements introduced in the male genome therefore can mobilize, and this causes genome instability in the F1 germline, causing atrophic gonads and sterility.

Wild-caught female (P strain) has P elements in her pi cluster and therefore piRNAs in her eggs that block transposition in the germline of the embryo. The F1 are fertile.

“Hybrid dysgenesis” explained:
piRNAs are present in P strain but not M strain

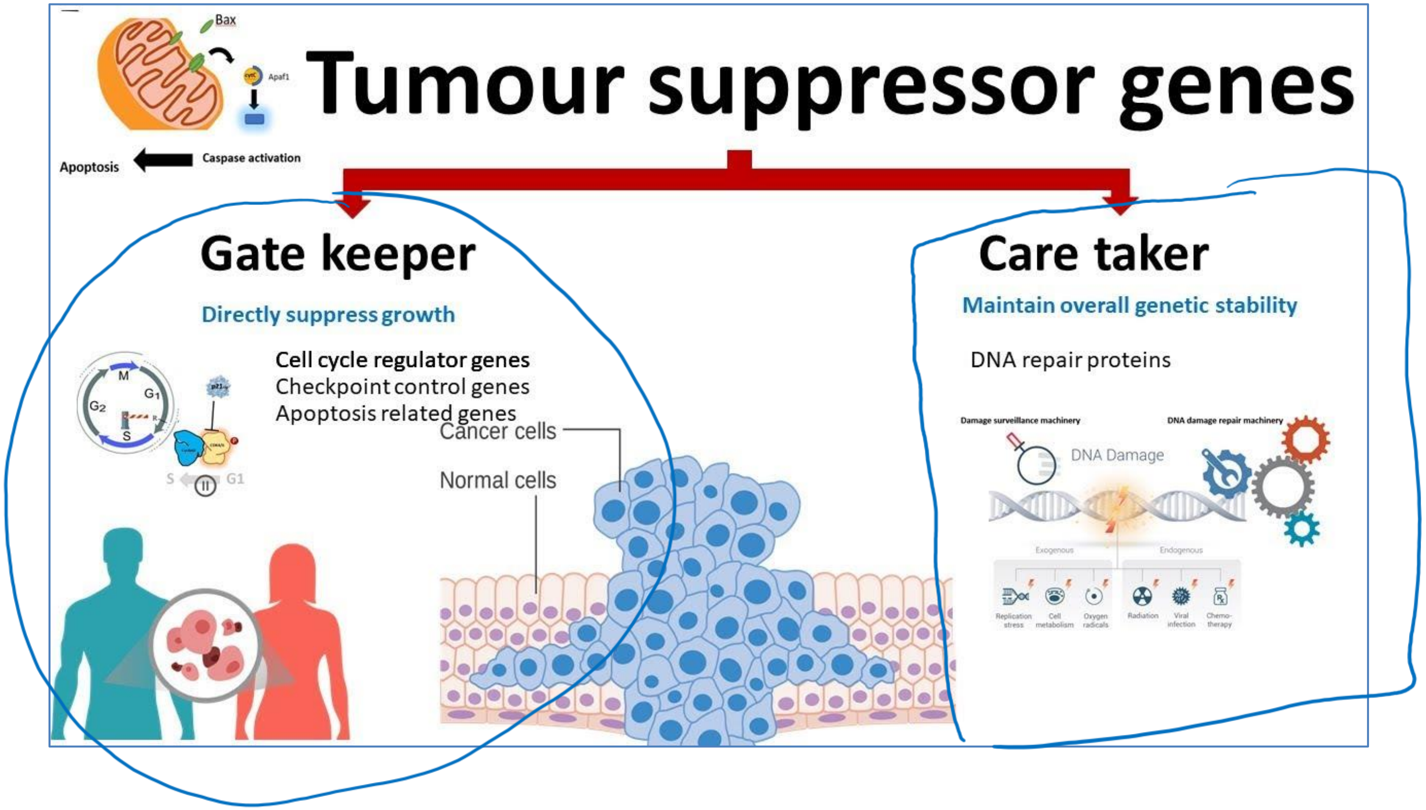




Griffiths et al., *Introduction to Genetic Analysis*, 12e, © 2020

Repression of transposable element mobility in humans may suppress tumor formation

The p53 gene is one of the most studied mutations in human cancer.
Mutations in p53 are found in most tumor types and contribute to the complex network of molecular events leading to tumor formation.
Hypothesis: p53 plays a role in repressing transposable element mobility.



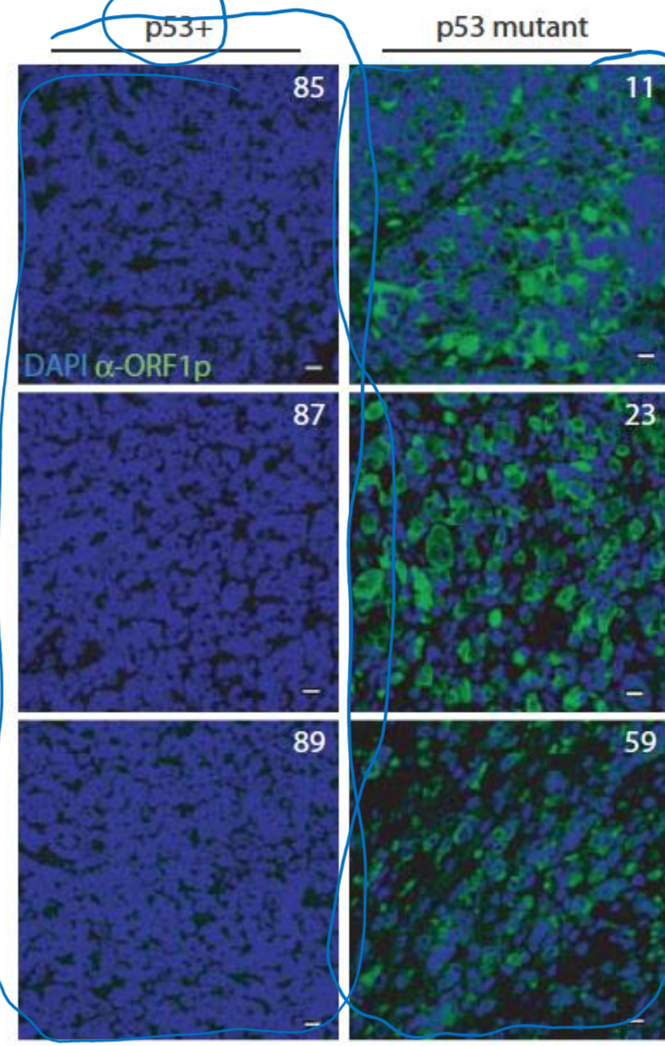
[p53 mutations in 10,000 cancer patients shed new light on gene's function -- ScienceDaily]

Wilms (kidney) tumors mutant for p53 show elevated human LINE1 ORF1p expression

blue = DNA

green = human LINE-1 ORF1p expression

Element	Transposition	Structure
LINEs	Autonomous	ORF1 ORF2 (*pol*) AAA
SINEs	Nonautonomous	AAA
DNA transposons	Autonomous	← transposase →
	Nonautonomous	← →

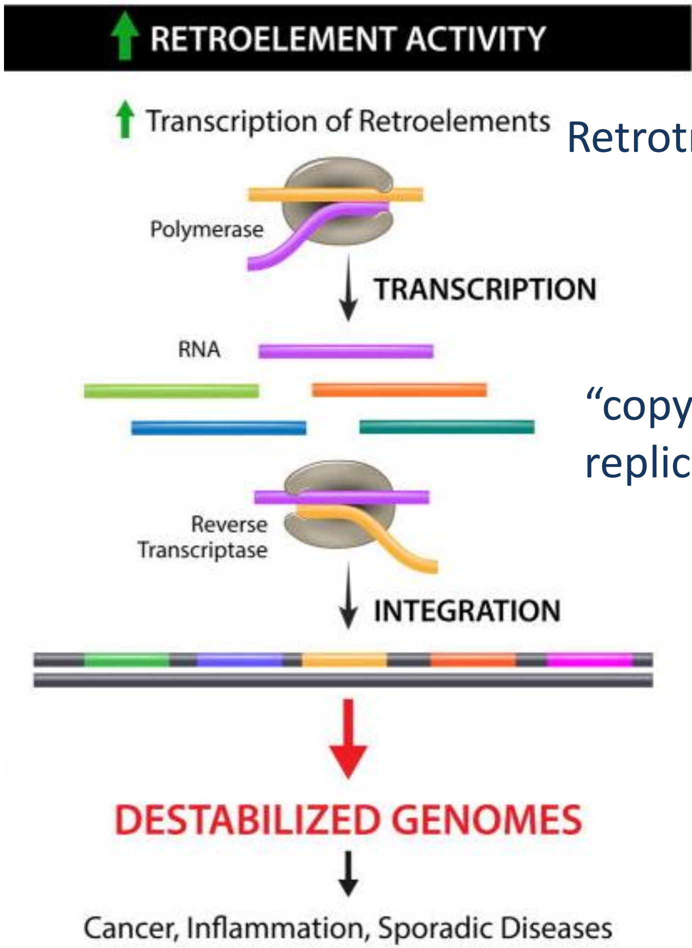


6 tumor samples

Deregulated retroelements stratify with p53 mutations in Wilms tumors. Compared with Wilms tumors that are wild type for p53 (*left* panels: 85, 87, and 89), Wilms tumors that are mutant for p53 (*right* panels: 11, 23, and 59) show dramatically elevated human LINE-1 ORF1p expression (α-ORF1p [green]; counterstained to visualize DNA [blue])

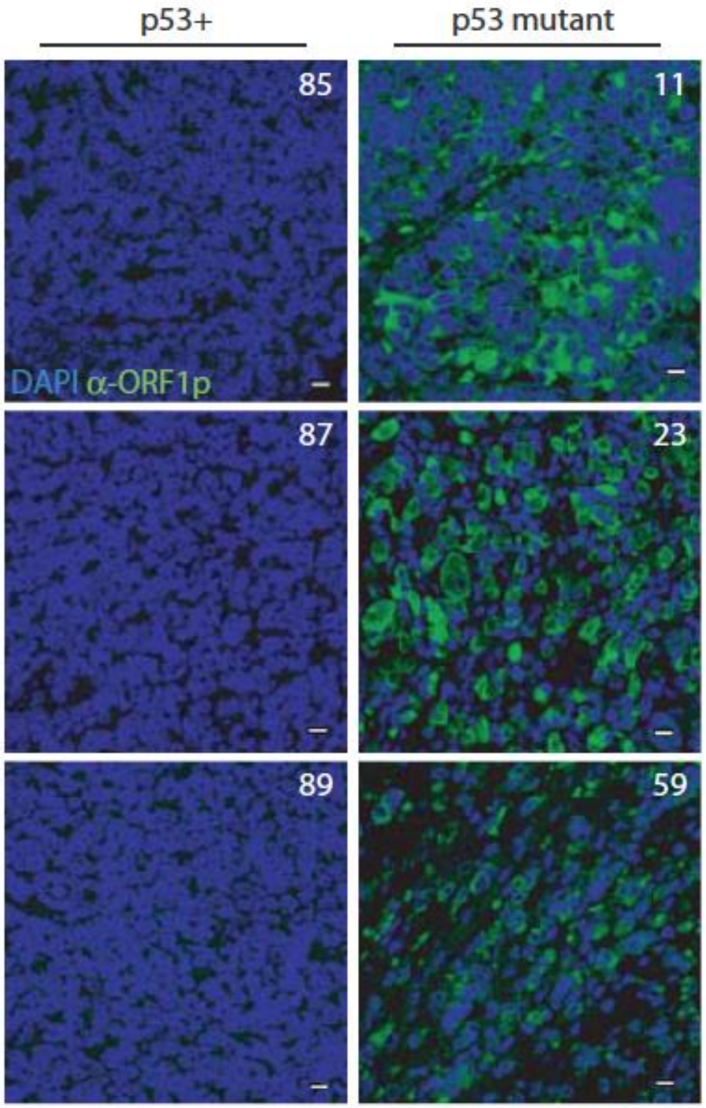
[p53 genes function to restrain mobile elements - PubMed (nih.gov)]

Hypothesis: Loss of p53 in kidney cells leads to deregulation of retrotransposon activity and to tumorigenesis.



Retrotransposons

"copy-paste" replication



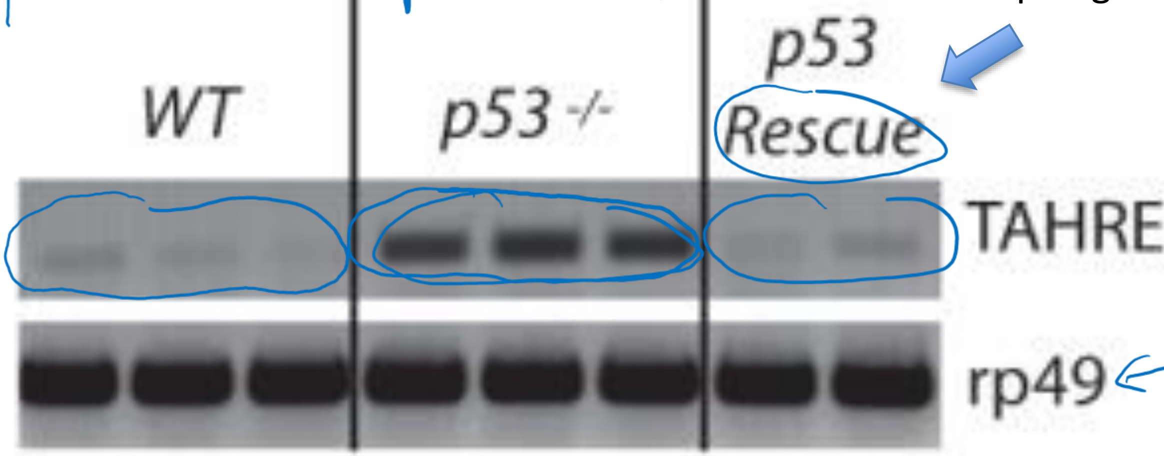
[p53 genes function to restrain mobile elements - PubMed (nih.gov)]

The p53 protein is required for silencing transposon mobility in *Drosophila*

does Dros p53 regulate TE mobility?

Flies that are p53-/- but with a wild type p53 gene added back

RNA transcripts from the TAHRE retrotransposon



a Drosophila retrotransposon

loading control

[p53 genes function to restrain mobile elements - PubMed (nih.gov)]

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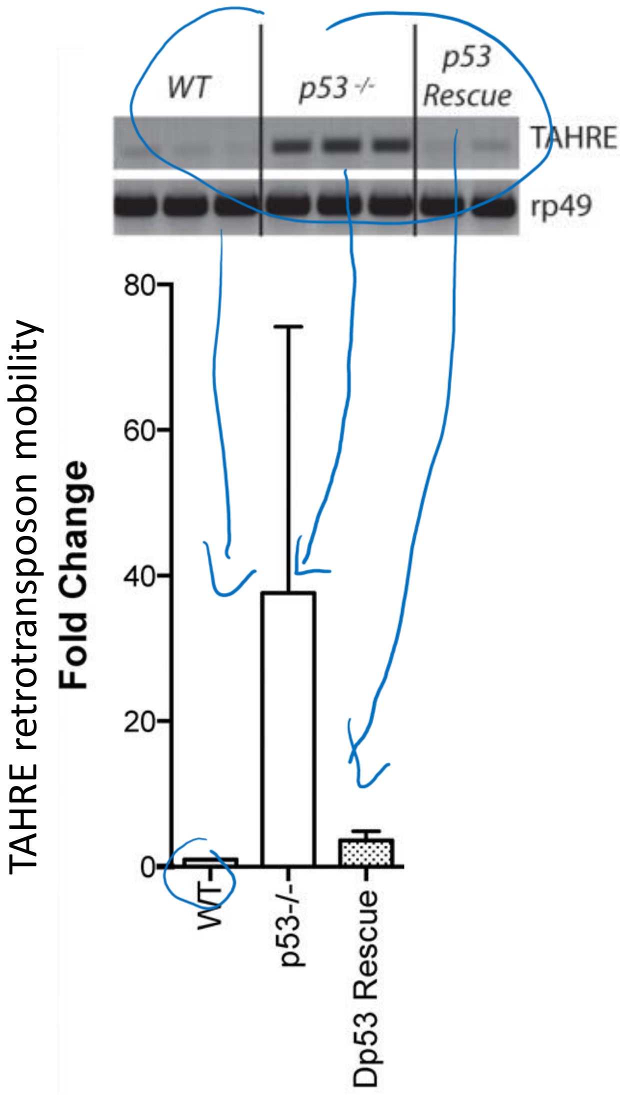
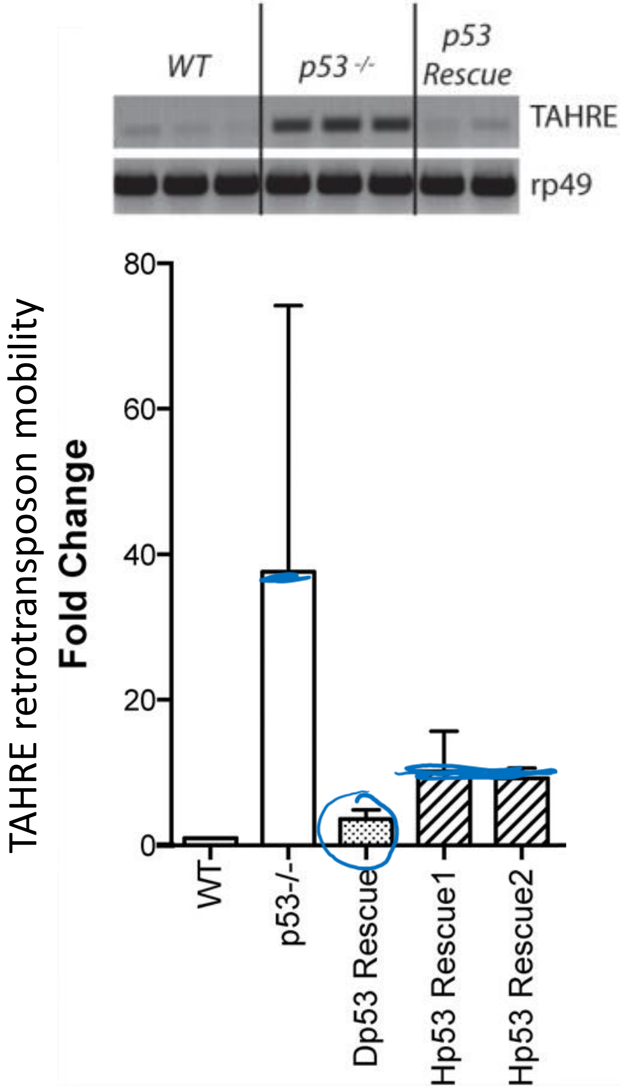


1. WT : wild type fly
2. p53-/- : P53 mutant fly
3. Dp53 Rescue: p53 mutant fly + transgenic wild type Drosophila p53

p53 genes function to restrain mobile elements - PubMed (nih.gov)

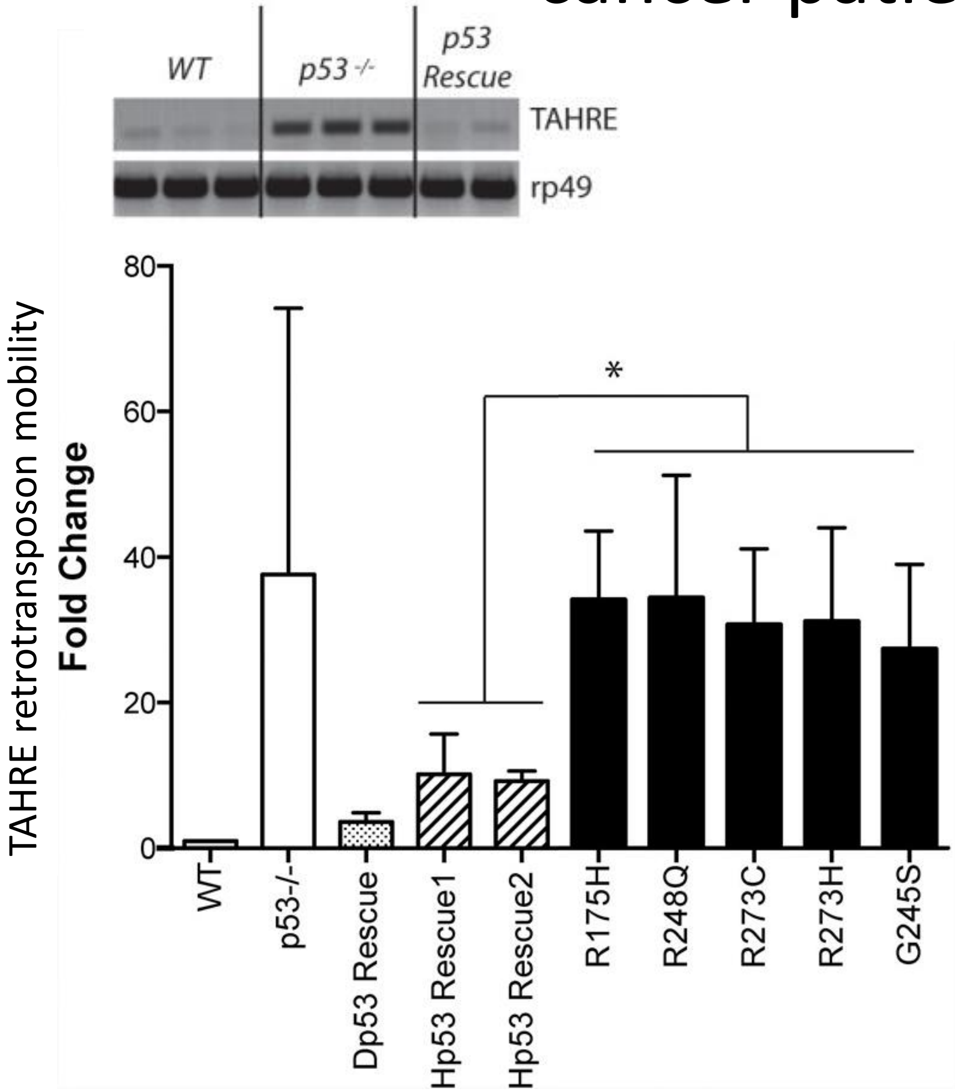
Human p53 can block retrotransposon activity in p53-mutant flies



1. WT : wild type fly
2. p53-/- : P53 mutant fly
3. Dp53 Rescue: p53 mutant fly + transgenic wild type Drosophila p53
4. Hp53 Rescue1: p53 mutant fly + transgenic wild type human p53
5. Hp53 Rescue2: p53 mutant fly + transgenic wild type human p53 (replicate)

[p53 genes function to restrain mobile elements - PubMed (nih.gov)]

Human p53 corrects dysregulated transposon activity in p53– flies, but p53 variants commonly seen in cancer patients do not.



1. WT : wild type fly
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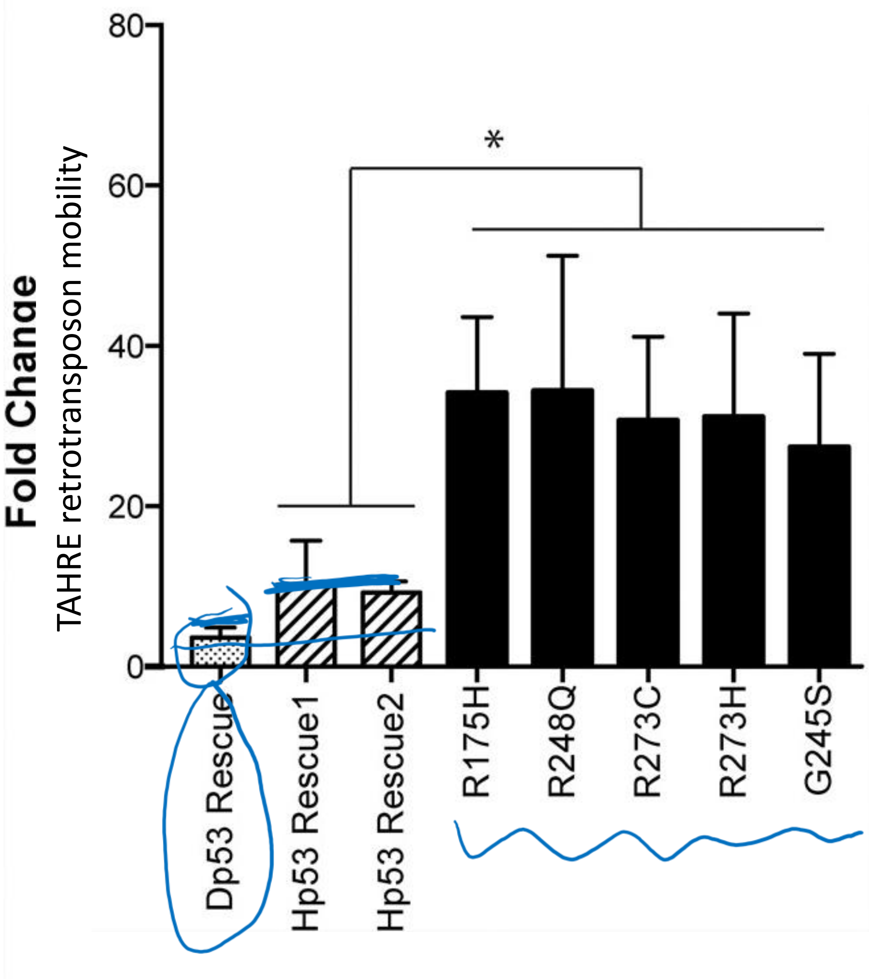
3. Dp53 Rescue: p53 mutant fly + transgenic wild type Drosophila p53
4. Hp53 Rescue1: p53 mutant fly + transgenic wild type human p53
5. Hp53 Rescue2: p53 mutant fly + transgenic wild type human p53 (replicate)
6. Black bars: p53 mutant fly + a transgenic human mutant p53 allele from a cancer patient

[p53 genes function to restrain mobile elements - PubMed (nih.gov)]

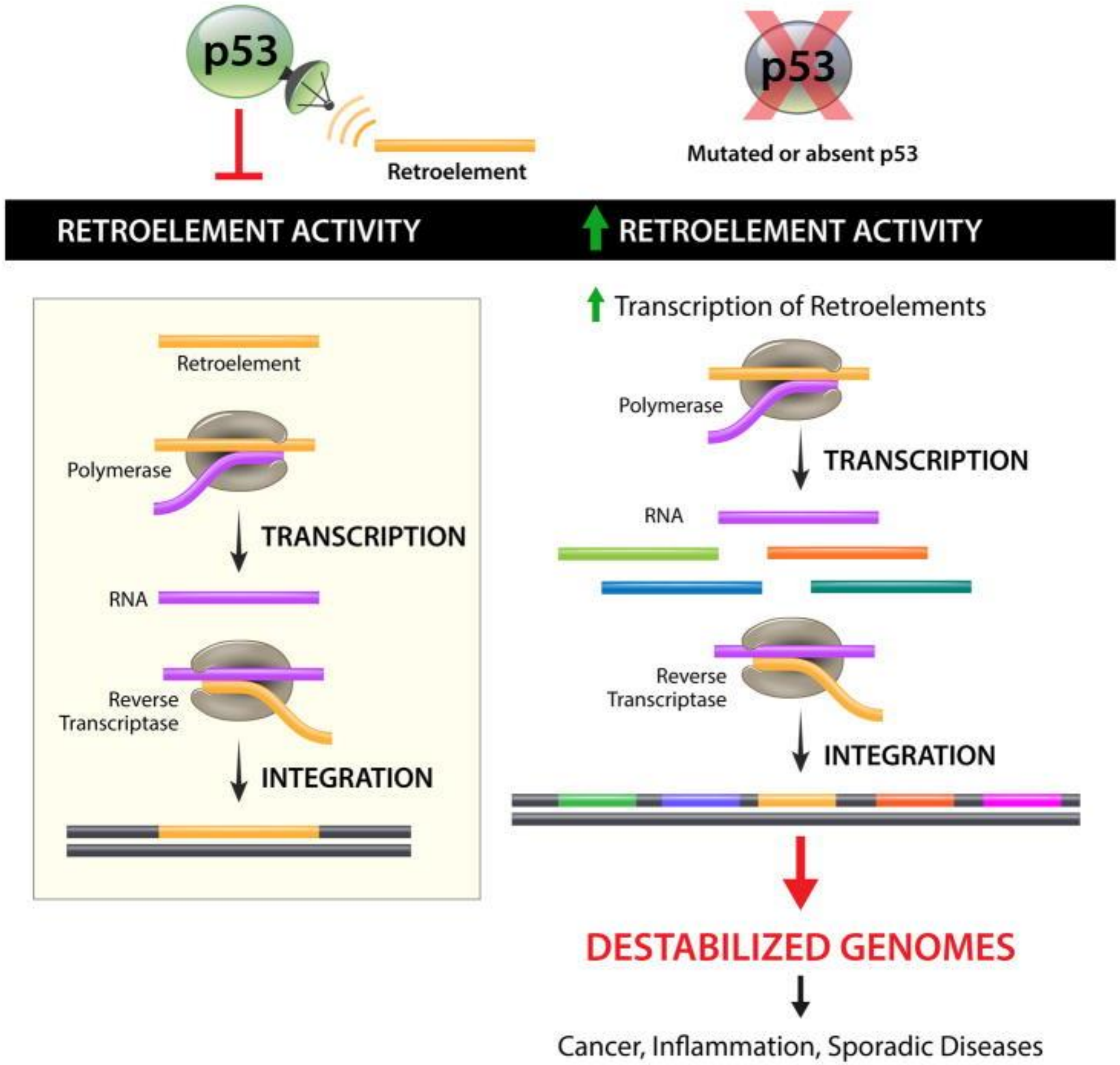
Think Break

Data Interpretation

Which strain has the most p53 activity?



The p53 protein promotes genome stability by silencing transposon mobility



[p53 in the Game of Transposons (nih.gov)]

DNA Mutations

Causes and Consequences

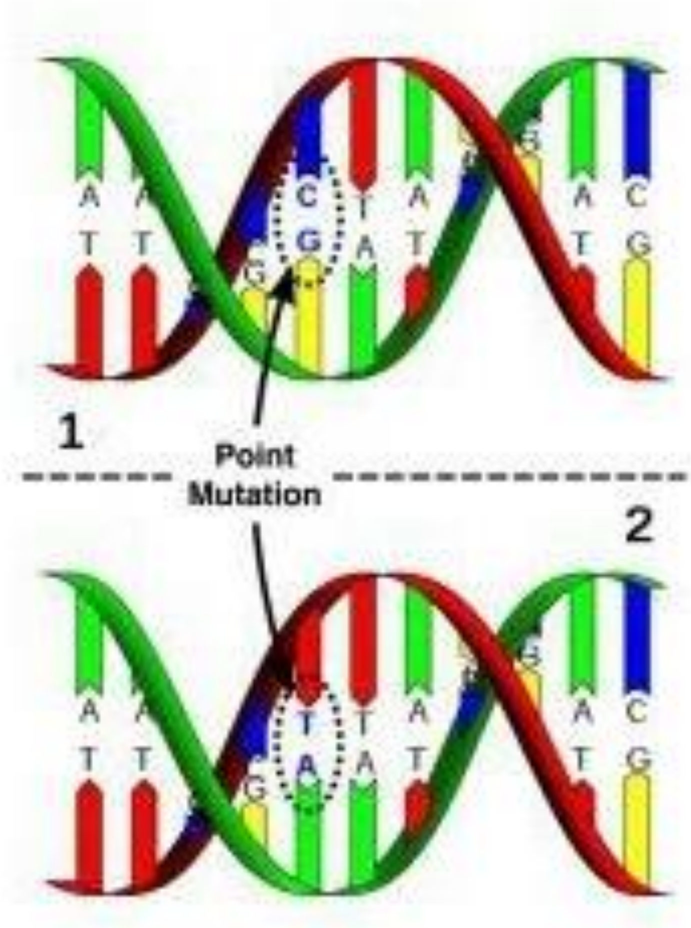
What is a mutation?

What are the consequences of mutations?

Where do mutations come from?

A mutation is a change in DNA sequence

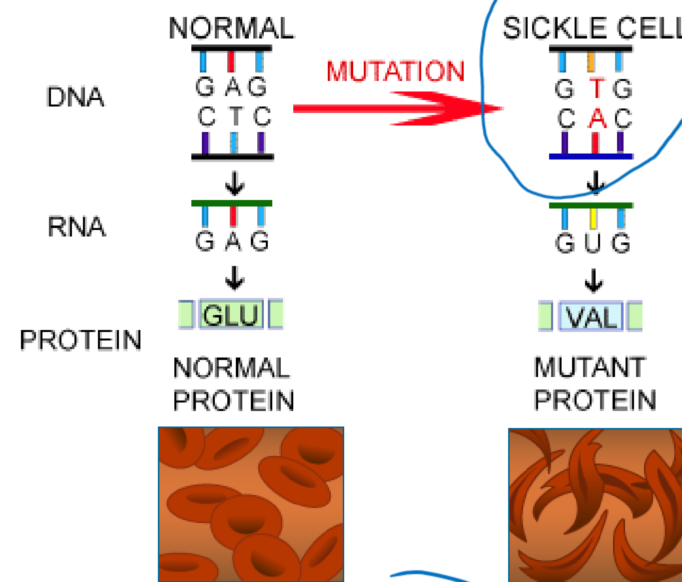
- Any unpredictable change in the structure or amount of the DNA of and organism is called a **mutation**.
- Most mutations occur in **somatic (body) cells** and are not passed on from one generation to the next.
- Only those mutations that occur in the formation of **gametes** can be inherited.

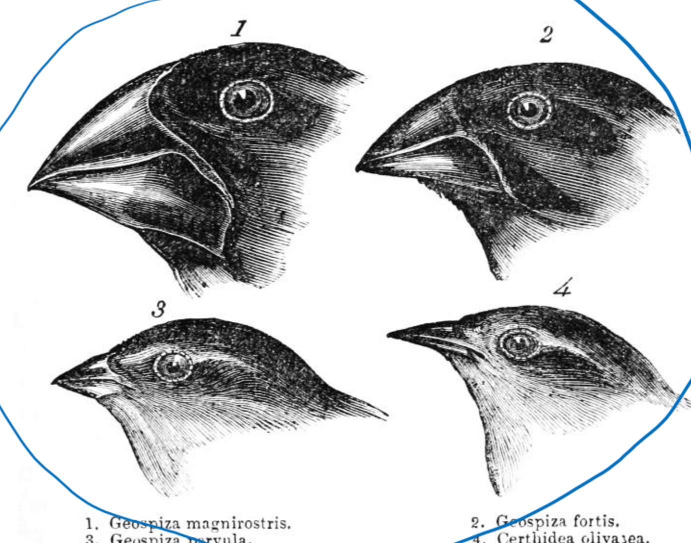


A mutation is a change in DNA sequence

<u>Mutations can:</u>

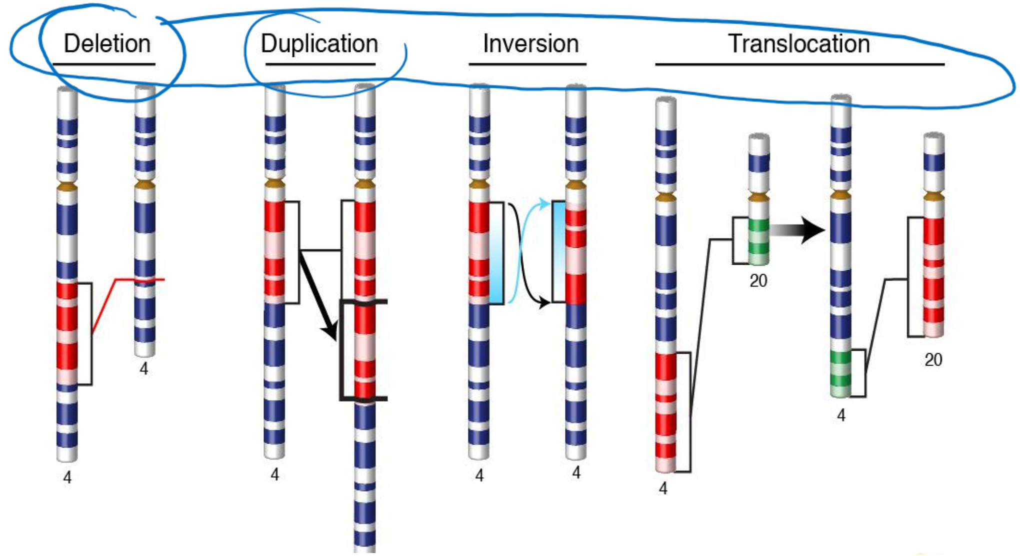
- be detrimental, neutral or beneficial
- affect coding sequences or regulatory sequences
- be an important source of genetic variation
    - Germline mutations are heritable.
    - Recombination during meiosis generates different combinations of alleles
    - This allows for adaptation/evolution



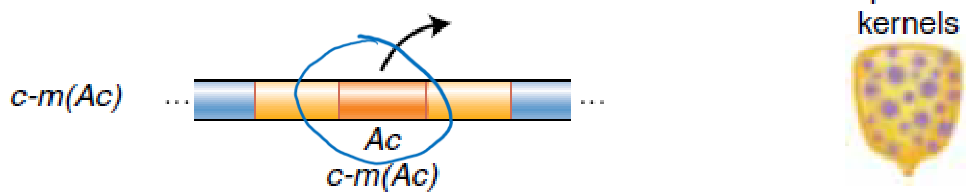


Mutations can occur on different scales

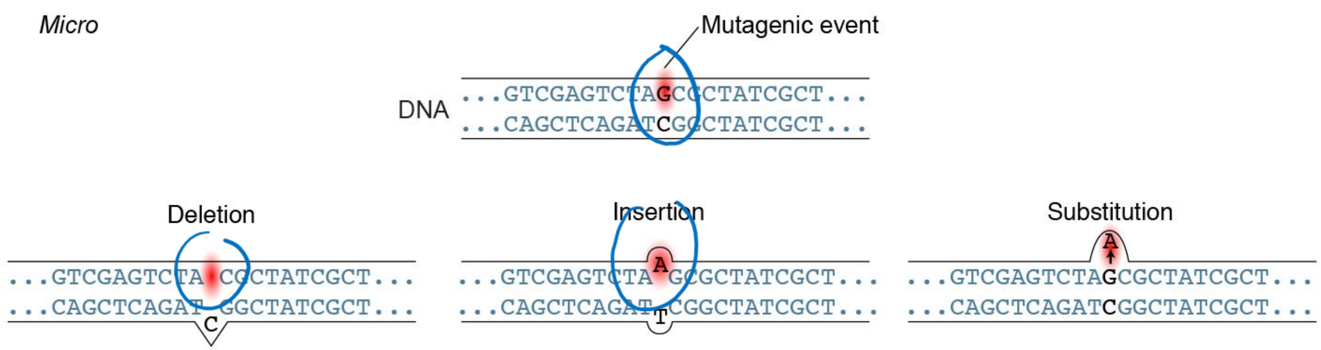
Chromosomal mutations: gain or loss of all or part of a chromosome



Insertional mutations: insertion of large regions of DNA, e.g. transposable elements

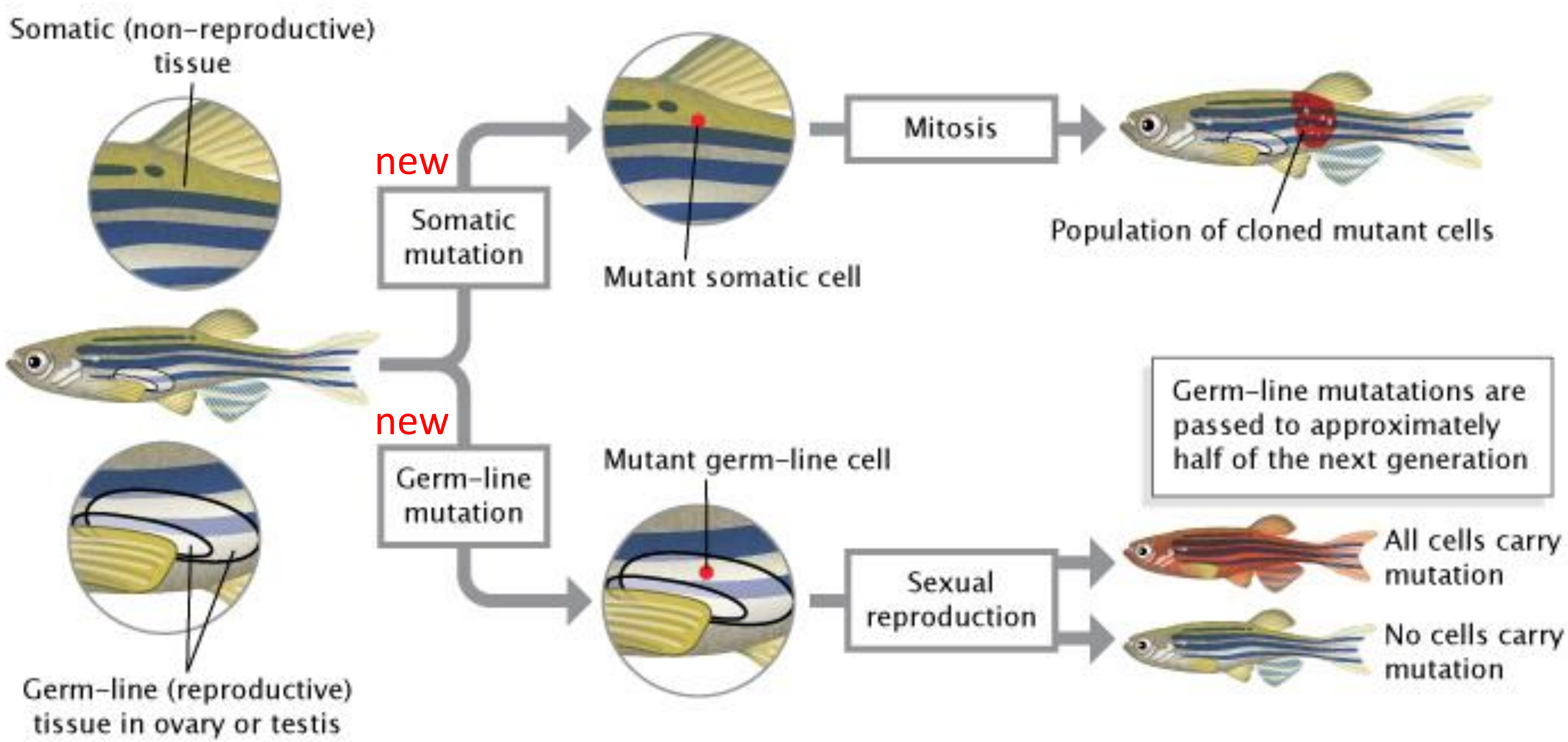


Point mutations: changes in a single nucleotide or addition/deletion of one or more nucleotides



[(genome.gov)]

Mutations in the germline, but not the soma, can be inherited



[Genetic Mutation | Learn Science at Scitable (nature.com)]

Think Break

Why will only half of the F1 individuals carry the mutation?